English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

Computer Science, 31 May 2026 Test 2

Problem 1. (0.75 points) Let (X, Y) have joint density

$$f_{X,Y}(x,y) = \begin{cases} cxy, & 0 < y < \sqrt{x} < 1, \\ 0, & \text{otherwise.} \end{cases}$$

1. (0.5 points) Find c, $\mathbb{E}[X]$ and $\mathbb{E}[X \mid Y = 1/2]$.
2. (0.25 points) Find the distribution (density) of $Z = Y/\sqrt{X}$.

Problem 2. (1 point) Let all $X_i, Y_i \sim U(0,1)$, $i \geq 1$, be independent and identically distributed. Set $Z_i = \mathbb{1}_{\{(X_i-1/2)^2+(Y_i-1/2)^2 \leq 1/4\}}$ and estimate π by $\widehat{\pi}_n = \frac{4}{n}\sum_{i=1}^{n} Z_i$.

1. (0.25 points) Calculate $\mathbb{E}[Z_i]$, $D[Z_i]$, $\mathbb{E}[\widehat{\pi}_n]$ and $D[\widehat{\pi}_n]$.
2. (0.25 points) Find the smallest n for which $\mathbb{P}(|\widehat{\pi}_n - \pi| < 0.01) \geq 0.95$.
3. (0.25 points) For $n = 10^6$, find the smallest $\delta > 0$ such that $\mathbb{P}(|\widehat{\pi}_n - \pi| < \delta) \geq 0.95$.
4. (0.25 points) By what factor must the number of trials in part 2 increase to reduce the error by a factor of 10 at the same 95% confidence?

Problem 3. (1 point) Let X, Y be independent $U(0,1)$ variables, and set $U = \max(X, Y)$ and $V = \min(X, Y)$.

1. (0.3 points) Find $f_{V,U}(v, u)$.
2. (0.3 points) Find $f_{V|U}(v \mid u)$ and $\mathbb{E}[V \mid U]$. Interpret the result.
3. (0.4 points) Define $Z = V/U$, the ratio of the smaller value to the larger. Find $f_{Z,U}(z, u)$. Are Z, U independent? What is the marginal distribution of Z?

Problem 4. (1 point) Two systems monitor a server for incidents. Its load is $\Theta \sim U(0,1)$. The first monitoring system is new and its detection time depends on the load: $X \mid \Theta = \theta \sim \text{Exp}(\theta)$. The second is old and calibrated, with detection time $Y \sim \text{Exp}(1)$. Conditional on a fixed Θ, X, Y are independent. The new system detects the incident first, so $X < Y$.

1. (0.3 points) Find $\mathbb{P}(X < Y \mid \Theta = \theta)$.
2. (0.4 points) Find the conditional density of $\Theta \mid X < Y$.
3. (0.3 points) Find $\mathbb{P}(\Theta > 1/2 \mid X < Y)$.

Problem 5. (0.85 points) Let X_1, X_2 be independent $N(0,1)$ variables, and $Y_1 = X_1/X_2$.

1. (0.25 points) Find $f_{Y_1}(y_1)$.
2. (0.25 points) Find $\mathbb{E}[Y_1]$.
3. (0.3 points) Show that $Z = 1/Y_1$ has the same distribution as Y_1 and find $\mathbb{P}(Y_1 > 1)$.